

# Agentic Delegation

and the Language Frontier

Quispe & Xu (2026), arXiv:2605.25438v2

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# 1. Citation audit: the discrepancy is real

**Current v2:** Alexander Quispe and Kevin Xu, “Agentic Delegation and the Language Frontier of Software Developers: A Model and Evidence from Claude Code on GitHub,” July 8, 2026.

## Earlier circulation

“Coding Beyond Your Training: Claude Code and the Technological Frontier of Software Developers,” Alexander Quispe only.

## Correct status

Preliminary, unrefereed **arXiv economics preprint**; not NBER. v1: May 25. v2: July 7; PDF dated July 8.

## 2. Question, claim, and non-claim

### Research question

Does agentic AI expand the languages in which a developer can ship working code, beyond accelerating work in familiar languages?

**Claim:** the developer's observed **production frontier** expands.

**Non-claim:** the developer learns to code unassisted in the new language. Delegated output remains joint production: the human selects, specifies, verifies, and ships.

**Unit:** developer  $i \times$  language  $k \times$  month  $t$ .

### 3. Three capabilities

#### Solo $S$

Expected human match  $s\mu$ , activation cost  $b$ , and CARA risk penalty from match uncertainty.

#### Augmentation $C$

Adds  $\gamma s - r_C$ : suggestions are useful only with language-specific foothold  $s$ .

#### Delegation $D$

Agent executes share  $\lambda$ ; output uses general ability  $a$ , capability  $z(A)$ , verification cost  $\kappa$ , and agent-error risk.

One modelling choice drives the result: **AI supplies delegated execution, not merely a substitutable productivity input.**

## 4. The developer's choice is not effort

$$M^1 = \{S, C\}, \quad M^2 = \{S, C, D\},$$
$$V_{ikt}^g = \max_{m \in M^g} V_{ikt}^m, \quad Z_{ikt}^g = 1\{V_{ikt}^g \geq 0\}, \quad N_{it}^g = \sum_k Z_{ikt}^g.$$

### Economic decision

Given an opportunity, choose the mode with the highest certainty-equivalent surplus, then enter only if that surplus is nonnegative.

**Audit point:** there is no endogenous continuous effort variable. An “agent effort problem” belongs to a different paper/model.

## 5. Thresholds and the delegation advantage

$$T^S = b - s\mu + \frac{\rho s^2}{2\pi}, \quad T^1 = T^S - \max\{0, \gamma s - r_C\}.$$

For unfamiliar  $k$ , Assumption 1 imposes  $\gamma s - r_C \leq 0$ , so  $T^1 = T^S$ . Define

$$B \equiv T^S - T^D = \underbrace{\lambda[az(A) - s\mu]}_{\text{execution substitution}} - \underbrace{\kappa(a, s) - r_D}_{\text{verification + compute}} + \underbrace{\frac{\rho}{2} \left[ \frac{(2\lambda - \lambda^2)s^2}{\pi} - \sigma_D^2 \right]}_{\text{risk substitution}}.$$

### Condition

Delegation changes entry only when  $B > 0$ . Better verification technology can enlarge the band; it does not guarantee a positive band everywhere.

## 6. Propositions 1–2

**Proposition 1 (weak frontier expansion).**

$$M^1 \subset M^2 \implies Z_{ikt}^2 \geq Z_{ikt}^1 \implies N_{it}^2 \geq N_{it}^1.$$

The agent can always be ignored; strict expansion is not automatic.

**Proposition 2 (activation band).** For unfamiliar  $k$ , Assumption 1 and  $B > 0$  imply

$$Z_{ikt}^2 - Z_{ikt}^1 = 1\{T_{ikt}^D \leq \omega_{ikt} < T_{ikt}^1\}.$$

With continuous conditional CDF  $F$ , activation probability is  $F(T^1) - F(T^D)$ .

## 7. Proposition 3: stock and flow

If constant first-use hazards satisfy  $p_{ik}^2 \geq p_{ik}^1$ , then

$$\Delta C_i(h) = \sum_{k \in U_i} [(1 - p_{ik}^1)^{h+1} - (1 - p_{ik}^2)^{h+1}] \geq 0.$$

### Closed frontier

If  $p^1 = 0 < p^2$ , the cumulative gap grows strictly and concavely over the observed horizon.

### Endpoint/domain caveat

With  $p^1 > 0$ , monotonic growth needs the paper's additional “no catch-up” inequality;  $p^2 \geq p^1$  alone gives nonnegativity, not monotonicity.

# 8. Data construction

| Object                               | Size                          |                         |
|--------------------------------------|-------------------------------|-------------------------|
| Window                               | Jan 2024–Apr 2026 (28 months) |                         |
| Initial / estimation developers      | 5,825 / <b>5,346</b>          |                         |
| Treated / not-yet-treated control    | <b>2,813</b> / <b>2,533</b>   | <b>Treatment:</b> first |
| Developer-month rows                 | <b>149,688</b>                |                         |
| Commits / developer–repository pairs | 3.2m / 133k                   |                         |
| Changed files classified by Linguist | <b>57m</b>                    |                         |
| Claude-coauthored detection universe | 7.8m commits                  |                         |

machine-readable Claude co-author trailer. Outcomes use non-merge commits and programming-language files.

## 9. Empirical strategy

### Estimator

Callaway–Sant’Anna group-time ATT, doubly robust, not-yet-treated comparisons, one-month anticipation, varying base periods, 1,000 developer-clustered multiplier-bootstrap draws.

**Four frontier outcomes:** monthly distinct languages, newly used languages, cumulative languages, and Shannon entropy. Repositories and commits are diagnostics.

### Interpretation discipline

Conditional parallel trends handles staggered timing and observed pre-period differences. It does not handle a new-project shock that both induces Claude adoption and changes languages.

## 10. Main numerical results

| Outcome          | $e = 0$        | $e = 1$        | $e = 2$        | Simple ATT     | Active languages |
|------------------|----------------|----------------|----------------|----------------|------------------|
| Active languages | 2.528 (.063)   | 1.227 (.064)   | .693 (.067)    | 1.178 (.052)   |                  |
| Newly used       | 1.193 (.051)   | .126 (.034)    | -.018 (.038)   | .294 (.030)    |                  |
| Entropy          | .382 (.009)    | .189 (.011)    | .102 (.012)    | .172 (.010)    |                  |
| Cumulative       | 1.604 (.054)   | 1.892 (.063)   | 2.072 (.086)   | 1.930 (.068)   |                  |
| Repositories     | 1.494 (.058)   | .838 (.057)    | .449 (.057)    | .742 (.057)    |                  |
| Monthly commits  | 35.080 (2.085) | 21.046 (2.292) | 12.110 (2.609) | 18.677 (2.269) |                  |

roughly triple relative to the .90 pre-mean; new languages are nearly four times the .31 pre-flow.  
Cumulative languages have significant pre-trends and are descriptive.

## 11. Mechanism evidence: headroom

| Adoption-month newly used languages | Specialist          | Generalist   |
|-------------------------------------|---------------------|--------------|
| High pre-adoption activity          | <b>.981</b> (.084)  | .301 (.100)  |
| Low pre-adoption activity           | <b>2.388</b> (.099) | 1.015 (.092) |

### Matches Proposition 4's headroom margin

Within activity halves, narrow-portfolio specialists have more unfamiliar candidates and show much larger first-use effects.

The general-ability prediction is not cleanly supported: commit volume is a poor ability proxy and low-volume users show larger measured effects.

## 12. Robustness and what it cannot rule out

### Survives:

- ▶ removing first-Claude language;
- ▶ excluding all Claude commits;
- ▶ activity conditioning;
- ▶ stricter pre-activity filters;
- ▶ competing-agent screens;
- ▶ placebo dates.

### Still unresolved

A developer may start an unfamiliar-language project and adopt Claude because of that same project. The observed jump can combine treatment with selection on a time-varying shock.

**Correct label:** sharp, robust event-time association consistent with delegation.

## 13. Symbolic and handwritten audit

### SymPy result

#### Exact identity

```
simplify((T_s - T_d) - B) == 0
```

Rational example:  $T^D = -487/1250$ ,  
 $T^S = 81/100$ , and band width  
 $B = 2999/2500 > 0$ .

### Student action required

Add your own photo as  
hand/activation-band-  
derivation.jpg

Then recompile.

## 14. Proposition 3: paper $\rightarrow$ Lean $\rightarrow$ proof

**Paper statement.** If  $0 \leq p_k^1 \leq p_k^2 \leq 1$ , then  $\Delta C(s) = \sum_{k \in U} [(1 - p_k^1)^{s+1} - (1 - p_k^2)^{s+1}] \geq 0$ ; when  $U \neq \emptyset$  and  $p_k^1 = 0 < p_k^2 < 1$ , the gap is strictly increasing and concave.

**Lean translation and proof fragment.**

```
theorem dynamic_cumulative_language_effect :
  dynamic_cumulative_language_effectSpec := by
  intro Language _ unfamiliar p1 p2 hHazards
  dsimp; constructor
  •intro s -- weak gap:  $\Delta C(s) \geq 0$ 
    apply Finset.sum_nonneg; intro k hk
    have hkB := hHazards k hk
    have h0 :  $0 \leq 1 - p_2 k$  := by linarith
    have hle :  $1 - p_2 k \leq 1 - p_1 k$  := by linarith
    have := pow_le_pow_left0 h0 hle (s + 1)
    linarith
  •rintro ⟨hNe, hClosed⟩ -- closed frontier  $p_1 = 0 < p_2$ :
    -- strictMono_nat_of_lt_succ + pow_lt_pow_right_of_lt_one0 => strict growth + concavity
```

**Plain English.** A higher first-use hazard lowers every survival probability  $(1 - p)^{s+1}$ ; summing those nonnegative differences gives  $\Delta C(s) \geq 0$ . Lean checks this *exactly* under the paper's stated hypothesis  $p^2 \geq p^1$ ; the strict-increase and concavity half is proved on the closed-frontier benchmark  $p^1 = 0 < p^2$ , precisely the domain the proposition conditions on—no extra assumption is smuggled in.

### Verified build

lake build QX26AgenticDelegation: PASS (8318 jobs). Five of five selected endpoints compile with no sorry; EconCSLib -fast check also passed. Source-fidelity audits remain pending, so status is partial.

## 15. Verdict

The model cleanly shows how delegated execution can expand a **production** frontier through a lower entry threshold.

The data show the predicted activation spike, accumulation pattern, and specialist headroom signature. They do not causally separate delegation from project-driven adoption, and they do not show skill acquisition.

Paper, derivations, audit, and sources: <https://github.com/johnbarraza/ai-03-quispe>